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PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
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No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Acharya, Munjal Ph.D. (University of California, Irvine) Baulch, Janet Ph.D. (University of California, Irvine) Britten, Richard Ph.D. (Eastern Virginia Medical School) Nelson, Gregory Ph.D. (Loma Linda University) Parihar, Vipan Ph.D. (University of California, Irvine) Soltesz, Ivan Ph.D. (University of California, Irvine)		
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Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
Task Progress:	Overview of progress and administrative matters: The overall organization of the CNS NASA Specialized Center of Research (NSCOR) remains unchanged, where it is managed at the University of California, Irvine (UCI: PI & Director, Limoli) and is conducted between 3 other universities, including Stanford (PI and Co-director, Soltesz), Loma Linda University (LLU: PIs, Nelson & Vlkolinsky), and Eastern Virginia Medical School (EVMS: PI, Britten). The overarching hypothesis that links these studies seeks to define the mechanisms contributing to charged particle–induced disruptions to CNS functionality that impact risk over acute (mission critical) and more chronic (terrestrial) timeframes. The unique composition of the NSCOR team has begun to deliver some significant and unexpected findings (highlighted below), and is poised to embark on some exciting new studies implementing “Mixed Beam” paradigms, low dose rate studies with neutrons, Pion exposures in Switzerland, live animal imaging and recording at Stanford, PET scans at LLU and some higher-order executive function and problem solving paradigms linked to neuroproteomics at EVMS. Our studies have now begun to shed considerable light regarding cosmic radiation effects from the level of the synapse to multiple neurocognitive complications. Work using low LET (linear energy transfer) protons and He ions has uncovered some truly remarkable and persistent effects spanning a 1-year timeframe in which behavioral decrements are temporally coincident with long-range disruptions in circuit connectivity across multiple brain regions, changes transpiring amidst significant increases in neuroinflammation. Our first GCR (galactic cosmic radiation) simulations have uncovered some interesting behavioral decrements along with reductions in monoamine levels in the brain. Significant differences between the sexes highlights some new insights uncovered this last year. Work at Stanford has been coordinated between LLU and UCI and has linked reduced precision of place cell field recordings and neuronal ensembles to various charged particle exposures. These studies are also being coordinated with EVMS, where efforts are underway to understand how cosmic radiation exposure impacts Sharp Wave ripples and learning and memory consolidation. Work from EVMS continues to uncover an impressive array of executive function deficits at the inter-individual level in pre-trained rats and importantly, impaired capabilities for abstract reasoning uncovered by administering tasks of increasing complexity. Many of these changes have now been linked to layer specific deficits in long-term depression in the medial prefrontal cortex, changes that may also portend those changes seen in mice exhibiting reduced serotonin levels in the dorsal hippocampus. Lastly, additional work has been instrumental in defining two potentially useful countermeasures that inhibit adenosine kinase activity (acting through epigenetic mechanisms) and that inhibit colony stimulating factor 1 receptor (CSF1R) (acting through microglial depletion) that confer neurocognitive benefits under both radioprotection or mitigation settings. Research Highlights for the CNS NSCOR: Work under the NSCOR has been able to demonstrate direct links between charged particle induced disruptions to cognition and persistent damage to specific neuronal subsets interrogated by our behavioral tasks. In addition to our past years published findings, we have now published some interesting and unanticipated effects of cosmic radiation exposure that are not limited to microcircuits, but rather, extend to longer range disruptions in neurotransmission between distinct regions of the brain. Studies conducted over the course of 1 year between Stanford and UCI now show that connectivity between the hippocampus and perirhinal cortex are virtually abolished, revealing a remarkably persistent deficit that increases the neurocognitive risks associated with deep space travel. These results have now been published in Experimental Neurology. 1. Parihar, V.K., Maroso, M., Syage, A., Allen, B.D., Angulo, M.C., Soltesz, I. and Limoli, C.L. Persistent nature of alterations in cognition and neuronal circuit excitability after exposure to simulated cosmic radiation in mice. In Press, Exp. Neurol. (2018). We have also published a review article highlighting many of the adverse impacts of radiation and other flight stressors (weightlessness) in Surgical Neurology International. 2. Jandial, R., Hoshide, R., Waters, J.D and Limoli, C.L. Space brain: The negative effects of space exposure on the central nervous system. Surgical Neurol. Int. 9:9, doi: 10.4103/sni.sni_250_17 (2018). Other work extends many of our structural plasticity studies to ultra-high resolution electron microscopy, and demonstrates now for the first time how low dose charged particles reduced synaptic density and axonal myelination throughout the brain. These studies are under secondary review at the Journal of Comparative Neuroanatomy. 3. Dickstein, D.L., Talty, R., Bresnahan, E., Varghese, M., Perry, B., Janssen, W.G.M., Sowa, A., Giedzinski, E., Apodaca, L, Baulch, J.E., Acharya, M.A., Parihar, V. and Limoli C.L. Alterations in synaptic plasticity and myelination in response to exposure to high-energy charged particles. Submitted J. Comparative Neuroanatomy (2018).

	Work with rats between EVMS and LLU is progressing nicely and starting to reveal some interesting effects in long-term depression that is specific to layers (Layer 5) within the pre-frontal cortex. This correlates with both poor performance on the Attentional Set Shifting (ATSET) and increased mood disorders that has just recently been linked to depressed serotonin levels in the CA1 region of the hippocampus. Much of this work is currently underway and will also be linked to Dr. Britten's UCFlex assay – that interrogates higher-level problem solving known to be relevant for many aspects of astronaut performance. In summary, much of our ongoing studies have added to our knowledge of space radiation effects on the brain, and have now laid additional groundwork for some exciting upcoming studies utilizing updated mixed beam and GCR simulations, low dose rate studies with neutrons at Colorado State University, Pion exposures, in vivo recordings of specific neuronal ensembles, PET imaging, and neuroproteomics linked to epigenetic changes in the brain. Much of this work will also inform decisions on how best to implement ongoing and future countermeasure development in the CNS NSCOR.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Parihar VK, Maroso M, Syage A, Allen BD, Angulo MC, Soltesz I, Limoli CL. "Persistent nature of alterations in cognition and neuronal circuit excitability after exposure to simulated cosmic radiation in mice." Exp Neurol. 2018 Jul;305:44-55. Available online 11 March 2018. https://doi.org/10.1016/j.expneurol.2018.03.009 ; PubMed PMID: 29540322 , Jul-2018
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